



National University

of Computer & Emerging Sciences Peshawar Campus

Student Name: _____

Roll Number: _____

Semester: Fall 2025

Exam: Mid

Marks: 30

Weightage: 20 to 25

Time Allowed: 90 min

Date: 15, Oct, 2025

Course: CL-2001 Data Structure Lab

Instructor: M Saad Khan

Paper Type: Odd

Note:

- Attempt all questions.
 - Understanding the question is the part of Exam.
 - Create a folder with the format (RollNo_yourname). Include all the C++ files(23P0679_Q1.cpp) in that folder. Use the following format for naming the files (RollNo_QuestionNo.cpp).
 - An individual may be assigned a straight-forward 0 if the submitted assessed task is copied/cheated from another individual/internet.
 - Comments your Code Properly. Use meaning full variable names for each and every question.
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Problem: 1

Marks: 07

Create two arrays, array1 and array2, and find how many elements are common between them.

Note: Use only pointers to access array elements. Index-based solutions are not allowed — otherwise, zero marks will be awarded.

E.g : array1 {1, 2, 3, 4, 5} , array2 {3, 4, 5, 6, 7} output will be 3.

Problem: 2**Marks: 15**

Write a C++ program to simulate a Bus Boarding System using a queue implemented with arrays.

Your program should:

1. Ask the user how many passengers can wait in the queue.
2. Provide a menu with the following options:
 - Add a passenger to the queue (arrival).
 - Board the next passenger (remove from queue).
 - See who's next to board.
 - Display all waiting passengers.
 - Exit the program.
3. Ensure that the queue follows the FIFO (First In, First Out) principle.
4. Do not use any built-in queue or library functions.

Note: Passenger class has attributes name and age.

Problem: 3**Marks: 08**

Write a C++ program to delete a target number from a doubly linked list.

If the target node is located at an even index, then all nodes containing that target value should be deleted.

However, if the target node is located at an odd index, then only that specific node should be deleted.

E.g : If linked list is this 5 => 6 => 9 => 4 => 6 => 5 => 6 and target is 6 than output will be 5 => 9 => 4 => 5

If linked list is this 5 => 9 => 6 => 4 => 6 => 7 and target is 6 than output will be 5 => 9 => 4 => 6 => 7